

Research Methodology in CS (CSEG3060_3)

Lecture 9: Case Studies in IPR - landmark patent cases, licensing disputes, and software IP litigations

Unit 6: Patent Rights and New Developments in IPR • Mapped Course Outcome: CO5

Lecture 9: Case Studies in IPR — Landmark Patent Cases, Licensing Disputes, and Software IP Litigations

Course: Research Methodology in Computer Science (CSEG3060_3) **Program:** B.Tech (Computer Science) **Semester:** V **Unit:** 6 — Patent Rights and New Developments in IPR **Lecture Outcome Mapping:** CO5 — Analyze the legal, ethical, and institutional dimensions of intellectual property as they apply to research and innovation in computer science.

1. Introduction

Case law is the living architecture of intellectual property rights (IPR). While statutes such as the Indian Patents Act (1970), the U.S. Patent Act (35 U.S.C.), and the European Patent Convention (EPC) provide the textual foundation, the **contours** of patentability, the **scope** of licensing, and the **enforceability** of software-related rights are largely shaped by judicial interpretation. For a researcher in computer science, understanding these cases is not a peripheral legal exercise — it is central to understanding what can be protected, how conflicts are adjudicated, and what strategic considerations must inform technology transfer and commercialization.

This lecture surveys **three intersecting domains** of case law:

1. **Landmark Patent Cases** — decisions that redefined the boundaries of patentable subject matter.
2. **Licensing Disputes** — cases that clarified the doctrine of technology transfer, FRAND obligations, and compulsory licensing.
3. **Software IP Litigations** — cases that wrestled with the unique problems of protecting algorithmic, functional, and expressive elements of software.

The pedagogical objective is to equip students with the analytical capacity to read judicial reasoning, identify legal doctrines, and apply them to research-level decisions involving invention disclosure, licensing strategy, and software protection.

2. Conceptual Framework

Before entering the cases, three doctrinal frames must be established.

2.1 Patentability Thresholds

Across jurisdictions, patents are granted only for inventions that satisfy four classical requirements:

- **Novelty** — the invention is not part of the prior art.
- **Non-obviousness / Inventive step** — a person skilled in the art would not arrive at the invention through routine progression.
- **Industrial applicability / Utility** — the invention has a concrete, substantial, and credible use.
- **Statutory subject matter** — the invention falls within the categories of patentable subject matter defined by statute (e.g., "any new and useful art, machine, manufacture, or composition of matter" under 35 U.S.C. § 101; "a new product or process involving an inventive step and capable of industrial application" under Section 2(1)(j) of the Indian Patents Act).

The case law examined below grapples primarily with the **fourth threshold** — what kinds of subject matter can lawfully receive patent protection.

2.2 Licensing Doctrines

Licensing is the contractual mechanism through which patent rights are exploited without transferring ownership. The principal doctrinal axes are:

- **Express vs. implied license** — whether the licensee's authority arises from explicit contractual terms or from conduct of the parties.
- **Exclusive vs. non-exclusive license** — whether the licensee has sole commercial rights within a defined territory or field of use.
- **Compulsory license** — a state-imposed license granted where the patentee fails to work the patent, or where public interest demands access.
- **FRAND licensing** — the obligation to license standard-essential patents (SEPs) on Fair, Reasonable, and Non-Discriminatory terms.

2.3 Software IP: A Bifurcated Regime

Software is protected under a **dual regime** that varies by jurisdiction:

- **Copyright** protects the source and object code as a literary work, but not the underlying function or idea (per the idea-expression dichotomy).
- **Patent** protects novel and non-obvious technical implementations of algorithms, processes, and systems, subject to subject-matter eligibility.
- **Trade secret** protects proprietary code not disclosed to the public.

Cases in this lecture illustrate how each of these vehicles interacts — and collides — with software innovation.

3. Landmark Patent Cases

3.1 *Diamond v. Chakrabarty* (1980) — The Genesis of Modern Biotechnology Patents

Citation: 447 U.S. 303 (1980) **Forum:** U.S. Supreme Court **Parties:** Sidney Diamond, Commissioner of Patents and Trademarks v. Ananda Mohan Chakrabarty

Facts: Dr. Ananda Chakrabarty, a microbiologist working for General Electric, developed a genetically engineered strain of *Pseudomonas* bacteria capable of breaking down crude oil — a trait not found in naturally occurring bacteria. He filed a patent application covering the bacterium itself. The Patent Office denied the claim on the ground that living organisms are not patentable subject matter under 35 U.S.C. § 101.

Issue: Whether a living, man-made microorganism is a "manufacture" or "composition of matter" within the meaning of § 101.

Holding: In a 5–4 decision, the Supreme Court held that a live, human-made microorganism is patentable subject matter. Justice Burger wrote:

"Congress intended statutory subject matter to 'include anything under the sun that is made by man.'"

Reasoning: The Court drew a categorical distinction: "products of nature, whether alive or dead, are not patentable," but Chakrabarty's bacterium was "the product of human ingenuity" — it had "markedly different characteristics from any found in nature."

Significance:

- Established that biotechnological inventions are patentable so long as human intervention produces something not found in nature.

- Paved the way for patents on transgenic animals (*Harvard Oncomouse*, 1988), recombinant DNA, and stem-cell technologies.
- For computer science researchers in bioinformatics, it is the doctrinal ancestor of every software-implemented biological claim.

3.2 *Bilski v. Kappos* (2010) — Reaffirming the Boundaries of Abstract Ideas

Citation: 561 U.S. 593 (2010) **Forum:** U.S. Supreme Court **Parties:** Bernard Bilski and Rand Warsaw v. David Kappos

Facts: Bilski and Warsaw filed a patent on a method for hedging risk in commodities trading — essentially a series of mathematical steps performed by a human mind (with optional computer implementation). The USPTO rejected the claim as an abstract idea; the Federal Circuit, in *In re Bilski* (2008), had announced a "machine-or-transformation test" as the exclusive criterion for patent eligibility.

Issue: Whether the machine-or-transformation test is the sole test for § 101 patent eligibility.

Holding: The Supreme Court unanimously affirmed that the application was unpatentable as an attempt to patent an abstract idea, but **rejected** the Federal Circuit's attempt to make the machine-or-transformation test exclusive.

Reasoning: Justice Kennedy wrote the plurality opinion, reaffirming three historical exceptions to § 101 — **laws of nature, natural phenomena, and abstract ideas** — while declining to adopt a rigid, single test.

Significance:

- Preserved the possibility of business-method and software-related patents while tightening eligibility.
- Left open the methodology for assessing eligibility, which led to further Supreme Court intervention (see *Alice Corp.*).

3.3 *Mayo Collaborative Services v. Prometheus Laboratories* (2012) — The Two-Step Framework

Citation: 566 U.S. 66 (2012) **Forum:** U.S. Supreme Court

Facts: Prometheus Laboratories patented a method for calibrating drug dosages based on metabolite levels — a process that involved a law of nature (the relationship between drug metabolites and efficacy) coupled with conventional clinical steps.

Issue: Whether a claim that recites a law of nature coupled with routine steps is patent-eligible.

Holding: Unpatentable. The additional steps were "well-understood, conventional, and routine" and did not transform the unpatentable natural law into a patent-eligible application.

Reasoning: The Court announced a two-step framework:

1. Determine whether the claims at issue are "directed to" one of the patent-ineligible concepts (laws of nature, natural phenomena, abstract ideas).
2. If so, examine whether the additional elements "transform the nature of the claim" into a patent-eligible application.

Significance:

- Set the analytical template later deployed against software patents in *Alice Corp.*
- Researchers in computational biology, AI-based diagnostics, and personalized medicine must engineer their claims to include unconventional, technology-specific steps.

3.4 Association for Molecular Pathology v. Myriad Genetics (2013) — Genes and Products of Nature

Citation: 569 U.S. 576 (2013)

Facts: Myriad Genetics held patents on the BRCA1 and BRCA2 genes, isolated from human chromosomes, and on complementary DNA (cDNA) synthesized from them. The patents enabled Myriad to control access to diagnostic tests for breast and ovarian cancer risk.

Issue: Whether isolated human genes and cDNA are patent-eligible.

Holding:

- **Isolated DNA:** Not patentable. The genomic sequence is a "product of nature" even when extracted.
- **cDNA:** Patentable. cDNA is not naturally occurring; it is synthetically created through reverse transcription.

Significance:

- Demonstrated the court's continuing deference to the product-of-nature doctrine.
- Catalyzed a global debate over access to genetic data — particularly relevant to medical informatics researchers working with genomic datasets.

3.5 Alice Corp. v. CLS Bank International (2014) — The Software Patent Watershed

Citation: 573 U.S. 208 (2014) **Forum:** U.S. Supreme Court

Facts: Alice Corporation held patents covering a computerized method for mitigating settlement risk in financial transactions. The patents claimed (i) a method, (ii) a computer-readable medium, and (iii) a system — a common drafting strategy to capture software implementations across statutory categories.

Issue: Whether the claims recite a patent-eligible invention.

Holding: Unpatentable. The claims were "directed to" the abstract idea of intermediated settlement, and merely reciting "generic computer components" did not supply an "inventive concept."

Reasoning: The Court applied the Mayo framework:

1. The claims are directed to an abstract idea (a fundamental economic practice long prevalent in commerce).
2. The additional elements — generic computer implementation — are insufficient to transform the abstract idea into a patent-eligible application.

Significance: The **Alice/Mayo two-step test** remains the operative U.S. standard for software and business-method patent eligibility. Post-*Alice*, the USPTO's § 101 rejection rate for software claims rose dramatically. For computer science researchers, *Alice* mandates careful claim drafting: claims must recite **specific technical improvements** rather than abstract ideas implemented on generic hardware.

3.6 Indian Context: Novartis AG v. Union of India (2013)

Citation: Civil Appeal No. 2706 of 2013 **Forum:** Supreme Court of India

Facts: Novartis sought a patent for the beta-crystalline form of imatinib mesylate (marketed as Gleevec), a cancer drug. The Indian Patent Controller and the Intellectual Property Appellate Board (IPAB) rejected the application under Section 3(d) of the Indian Patents Act — a provision added in 2005 to prevent "evergreening" by requiring that new forms of known substances demonstrate significantly enhanced therapeutic efficacy.

Issue: Whether the new form of the known substance imatinib was patentable under Section 3(d).

Holding: The Supreme Court upheld the rejection. The new crystalline form did not demonstrate significantly enhanced efficacy over the known substance. Justice Aftab Alam wrote:

"The products of the invention are not of different efficacy. They are of the same efficacy as the products disclosed in the prior art."

Significance:

- Demonstrates India's explicit statutory resistance to evergreening, distinguishing it from U.S. doctrine.
- For pharmaceutical informatics and computational drug design researchers, the case signals that Indian patent law scrutinizes incremental optimization claims more strictly than U.S. law.

3.7 Bayer Corporation v. Natco Pharma (2012) — India's First Compulsory License

Citation: Compulsory License Application No. 1 of 2011 **Forum:** Controller General of Patents, Designs & Trade Marks (India)

Facts: Bayer held the Indian patent for sorafenib tosylate (Nexavar), a kidney and liver cancer drug. Natco Pharma applied for a compulsory license, citing Bayer's failure to make the drug available at a reasonably affordable price and its insufficient working of the patent in India.

Issue: Whether the requirements for a compulsory license under Section 84 of the Indian Patents Act were met.

Holding: The Controller granted the license, holding:

- **Reasonable requirements of the public** were not satisfied (drug priced at Rs. 2.8 lakh/month against Indian GDP per capita).
- **The patented invention was not worked in India.**
- The patentee was importing rather than manufacturing locally.

Significance:

- India's first granted compulsory license.
- Established a precedent that affordability constitutes "reasonable requirements of the public."
- For researchers, this case is foundational to understanding technology transfer mechanisms in access-to-medicine contexts.

4. Landmark Licensing Disputes**4.1 Microsoft Corp. v. Motorola, Inc. (2012) — The FRAND Standard**

Citation: No. C10-1823JLR (W.D. Wash. 2013); affirmed in part by the Federal Circuit **Forum:** U.S. District Court for the Western District of Washington

Facts: Motorola held patents declared essential to the H.264 video codec and the 802.11 Wi-Fi standard — both industry standards implemented in Microsoft's Xbox and Windows products. Microsoft sought a declaratory judgment on the FRAND obligations Motorola owed as a standard-essential patent (SEP) holder.

Issue: What terms constitute a "fair, reasonable, and non-discriminatory" royalty for SEPs?

Holding: Judge Robart determined that a reasonable royalty for SEPs must be calculated with reference to the **value of the patented technology's contribution to the standard**, not the overall value of the standard-compliant product. The court also assessed the **Georgia-Pacific factors** (15 factors developed in *Georgia-Pacific Corp. v. United States Plywood Corp.*, 1970, used to calculate reasonable royalties).

Significance:

- Anchored the **smallest salable patent-practicing unit (SSPPU)** approach to FRAND royalty calculation.
- For computer science researchers working on wireless, video, networking, and standardization, the case illustrates the commercial stakes of contributing to standards bodies.

4.2 Ericsson Inc. v. Lenovo Group Ltd. (2024) — Continuing FRAND Saga

Facts: Ericsson sued Lenovo over the licensing of SEPs essential to the 5G, Wi-Fi, and H.264 standards. The dispute concerned the appropriate FRAND rate and whether Lenovo was entitled to a license.

Significance:

- Demonstrates that FRAND remains a contested frontier in the 5G era.
- Reinforces that global FRAND disputes increasingly require coordinated resolution across jurisdictions.

4.3 Koninklijke Philips N.V. v. CD-Manufacturers (Indian Cases)

Philips has pursued enforcement against manufacturers of blank CDs and DVDs on the basis of patent and trade-mark rights. Indian courts have issued dynamic injunction orders requiring disclosure of customer details and payment of damages.

Significance:

- Shows how SEP holders can leverage Indian litigation to manage technology adoption.
- Highlights the procedural innovation of **dynamic injunctions** in IP enforcement.

4.4 Smith International v. Hughes Tool Co. (Fed. Cir. 1983) — Implied License Doctrine

Facts: Smith manufactured drill bits that, when operated, infringed Hughes Tool's patented drill bit design. Smith had previously purchased a license from a third party. The Federal Circuit held that Smith was an implied licensee because Hughes had consented to the manufacture and sale of the licensed tools.

Significance:

- Establishes that an implied license can arise from conduct, even without formal contractual language.
 - For researchers participating in technology-transfer negotiations, the case illustrates the latent risks of incomplete contractual documentation.
-

5. Software IP Litigations

5.1 Lotus Development Corp. v. Borland International (1996) — The Menu Command Wars

Citation: 49 F.3d 807 (1st Cir. 1995), affirmed by an equally divided Supreme Court (1996)

Facts: Lotus sued Borland for copying the Lotus 1-2-3 menu command hierarchy in Borland's Quattro and Quattro Pro spreadsheets. Borland conceded that the menu structure was not literally copied but argued that the hierarchy was a "method of operation" excluded from copyright protection under 17 U.S.C. § 102(b).

Issue: Whether the user interface of a software program — specifically, the hierarchical menu structure — is copyrightable.

Holding: The First Circuit held that the menu structure was a "method of operation" and not copyrightable. However, the Supreme Court affirmed by an equally divided court, leaving the issue unresolved as binding precedent.

Significance:

- Demonstrates that functional interface elements lie at the boundary of copyright.
- Influenced later rulings on look-and-feel litigation, including *Apple v. Microsoft* and *Apple v. Samsung*.

5.2 Apple Computer, Inc. v. Microsoft Corp. (1994) — The Look-and-Feel Settlement

Facts: Apple sued Microsoft over the graphical user interface of Windows, alleging that elements such as overlapping windows and trash icons infringed Apple's copyrights. Microsoft prevailed on

summary judgment, and the case settled.

Significance:

- Confirmed that GUI elements that are necessary to functional operation are not copyrightable.
- Set the stage for the proliferation of look-and-feel litigation in the late 1990s.

5.3 Oracle America, Inc. v. Google Inc. (2021) — The API Copyright Question

Citation: 141 S. Ct. 1183 (2021) **Forum:** U.S. Supreme Court

Facts: Oracle sued Google for copying the structure, sequence, and organization (SSO) of Java application programming interfaces (APIs) when developing its Android operating system. Google had copied approximately 11,500 lines of declaring code (the method headers and class declarations) but had written its own implementing code.

Issue: Whether the declaring code and the SSO of an API are copyrightable, and whether Google's use constituted fair use.

Holding: The Supreme Court assumed (without deciding) that the declaring code and SSO were copyrightable but held that Google's use constituted **fair use**. Justice Breyer's opinion emphasized:

- Google's commercial but transformative purpose.
- The limited amount of code copied.
- The fact that the use created a new platform (Android) that did not substitute for Java.

Significance:

- The most consequential software copyright decision in two decades.
- For software researchers, it clarifies that **functional compatibility** is a powerful fair-use defense.
- Has been read as supporting interoperability, but its boundaries remain contested.

5.4 McKernan v. Burex (UK — Software Patent Eligibility)

Facts (summary): The UK's Patents Act 1977 excludes "programs for computers ... as such" from patentability. McKernan v. Burex (later appeals and the *Aerotel/Macrossan* framework) established the test for assessing whether a computer-implemented invention is patentable as more than a "program for a computer as such."

The Aerotel/Macrossan four-step test (UK):

1. Properly construe the claim.
2. Identify the actual contribution.
3. Ask whether the contribution falls within excluded subject matter.
4. Check whether the contribution is technical in nature.

Significance:

- The UK and EPO apply a "technical character" test, in contrast to the U.S. Alice/Mayo framework.
- Computer science researchers targeting European patents must engineer claims to recite technical effects at each step.

5.5 Indian Cases: Microsoft Corporation v. Yogendra Pande (2005) and Rajah & Lata (Software Piracy)

Microsoft has pursued aggressive anti-piracy enforcement in India. The landmark *Microsoft v. Yogendra Pande* case resulted in a criminal conviction for software piracy, demonstrating the integration of criminal sanctions under the Indian Copyright Act, 1957.

Significance:

- Illustrates that software IP enforcement in India operates through both civil and criminal mechanisms.
- For Indian researchers, it underscores the personal liability risk associated with unauthorized software deployment in academic settings.

6. Case Study: Comparative Analysis

The following table consolidates the doctrinal posture across the principal cases discussed.

Case	Year	Jurisdiction	Doctrine at Stake	Computer-Science Relevance
Diamond v. Chakrabarty	1980	USA	Patentable subject matter	Biotech / bioinformatics
Bilski v. Kappos	2010	USA	Abstract ideas	Business-method software
Mayo v. Prometheus	2012	USA	Laws of nature	Computational diagnostics
Myriad Genetics	2013	USA	Products of nature	Genomic algorithms
Alice Corp. v. CLS Bank	2014	USA	Abstract ideas	All software patents
Novartis v. Union of India	2013	India	Section 3(d) — evergreening	Drug informatics
Bayer v. Natco	2012	India	Compulsory licensing	Pharma technology transfer
Microsoft v. Motorola	2013	USA	FRAND	Wireless / video standards
Oracle v. Google	2021	USA	Software copyright & fair use	APIs and interoperability
Lotus v. Borland	1996	USA	Look-and-feel	GUI design

7. Practical Implications for Computer Science Researchers

7.1 Invention Disclosure

Before publication or open-source release, researchers must conduct an invention disclosure assessment. Even an academic paper can establish prior art that prevents later patenting.

7.2 Claim Drafting for Software

In the post-Alice environment, patent claims directed at software should:

- Recite a specific technical improvement (e.g., "reduced memory consumption by X%").
- Avoid purely functional or business-method language.

- Include unconventional algorithmic steps.
- Specify the technical architecture (e.g., distributed, parallel, or hardware-accelerated).

7.3 Licensing Strategy

Researchers engaged in technology transfer must:

- Distinguish **patent license** from **copyright license** from **trade-secret license**.
- For SEPs, anticipate FRAND obligations before contributing to standards bodies.
- For sponsored research, negotiate **march-in rights** and **background IP terms**.

7.4 Software IP in Educational Institutes

Under Indian law and institutional policies:

- Software developed using institutional resources is typically owned by the institute.
- Student-authored software is governed by the institute's IP policy — typically requiring disclosure and offering royalty-sharing.
- Open-source licensing (GPL, MIT, Apache) interacts with copyright but **does not extinguish** patent rights unless explicitly waived.

8. IPR in Educational Institutes: Case Examples

8.1 The *Madey v. Duke University* (2002) Case

Citation: 307 F.3d 1351 (Fed. Cir. 2002)

Facts: John Madey, formerly a Duke University researcher, sued Duke for patent infringement after his lab equipment was used in research activities that fell outside the traditional experimental-use exception.

Holding: The Federal Circuit significantly narrowed the experimental-use defense, holding that **even non-profit research conducted with state-of-the-art facilities for commercial purposes is not protected**.

Significance:

- Universities and research institutions are treated as commercial actors for patent purposes.
- Educational institutes must establish robust IP policies, including invention assignment clauses, disclosure protocols, and technology-transfer offices.

8.2 Indian Institutional Policies

Indian institutions such as IITs, IISc, and BITS Pilani have adopted formal IP policies modelled on the U.S. Bayh-Dole Act (1980). The salient features include:

- Faculty ownership of copyright in scholarly works.
- Institute ownership of patents arising from sponsored research.
- Revenue-sharing models (typically 30–40% to inventors).

These policies have produced landmark technologies (e.g., the IIT Madras low-cost cancer screening device, the IISc supercomputer initiative) and demonstrate how IP frameworks catalyze innovation when properly structured.

9. Summary

The lecture has traversed more than four decades of case law to demonstrate that intellectual property is a dynamic, contested, and computationally consequential field. Three takeaways stand out:

1. **Patent eligibility for software and biotechnology remains volatile.** The Alice/Mayo framework continues to generate litigation, and the U.S. Supreme Court has not revisited the issue since 2014. Researchers must monitor ongoing developments, including pending cases and USPTO guidance (e.g., the 2019 Revised Patent Subject Matter Eligibility Guidance).
2. **Licensing disputes increasingly involve standard-essential patents.** The FRAND doctrine, shaped by *Microsoft v. Motorola* and its progeny, affects every researcher participating in standards bodies. Strategic licensing decisions — including the choice of jurisdiction, the structure of royalty stacks, and the negotiation of portfolio licenses — determine commercial viability.
3. **Software protection requires a multi-vehicle strategy.** Copyright protects expression; patents protect technical implementations; trade secrets protect confidentiality. Each has limitations, and the optimal portfolio depends on the nature of the innovation.

For computer science researchers, the operative principle is not to avoid IPR but to **integrate IP analysis into the research workflow** — from initial disclosure, through publication, to commercialization.

10. Exam Preparation and Practice Questions

10.1 Practice Questions

Short Answer (2–3 marks each):

1. State the four-step test announced in *Mayo v. Prometheus* and explain its application in *Alice Corp. v. CLS Bank*.
2. Distinguish between an "exclusive license" and a "compulsory license" with reference to the Indian Patents Act.
3. What is the FRAND doctrine? Identify one case where it was adjudicated.
4. Explain the significance of Section 3(d) of the Indian Patents Act and its application in *Novartis v. Union of India*.
5. Why was *Madey v. Duke University* significant for academic research?

Long Answer (10–15 marks each):

6. Critically analyze the U.S. Supreme Court's evolving stance on software patent eligibility from *Bilski* to *Alice Corp.*. How does this compare with the European "technical character" approach?
7. Discuss the licensing disputes surrounding standard-essential patents. With reference to *Microsoft v. Motorola* and *Ericsson v. Lenovo*, evaluate the doctrine of FRAND licensing.
8. Examine the doctrinal tension between copyright and patent protection for software, with reference to *Oracle v. Google* and *Lotus v. Borland*. What is the optimal IP strategy for an open-source software project?
9. Analyze the implications of the *Bayer v. Natco* compulsory license for technology transfer in pharmaceuticals. Can the same logic apply to computer-implemented inventions in India?
10. "Educational institutes are no longer islands of experimental use." Critically evaluate this statement in light of *Madey v. Duke University* and the IP policies of Indian institutions.

10.2 Analytical Hints

- Frame answers using the **issue–rule–application–conclusion (IRAC)** structure.
- When comparing jurisdictions, always anchor the comparison in **specific statutory provisions** (e.g., 35 U.S.C. § 101 vs. Section 3(d) of the Indian Patents Act).
- For technology-transfer questions, address both **licensing** and **assignment** as alternatives.

- For software-specific questions, distinguish between **copyright**, **patent**, and **trade secret** protection, and identify which vehicle is most appropriate for the given subject matter.

10.3 Suggested Reading

- *Intellectual Property: The Law of Trademarks, Copyrights, Patents, and Trade Secrets* — Bouchoux (Cengage).
 - *Software and Internet Law* — Mark A. Lemley et al. (Aspen).
 - Indian Patent Act, 1970 (as amended in 2005).
 - WIPO Intellectual Property Handbook (<https://www.wipo.int>).
 - USPTO Manual of Examining Procedure (MPEP), Chapter 2106 (Patent Eligibility).
-

11. Concluding Note

Case studies are not historical curiosities — they are the operational vocabulary of innovation policy. Whether the question is whether a CRISPR algorithm is patentable, whether an FRAND royalty can be extracted for 5G SEPs, or whether a university owns the copyright in student-developed software, the answer lies in the doctrinal evolution traced through cases like those examined in this lecture. Mastery of these cases distinguishes the technically informed researcher from the merely technically competent one — and in a field where research outputs are increasingly commercial assets, that distinction matters.